

Stellarium

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Astronomy is a hobby that really requires you to go out and observe the night sky but there are programs made to ease understanding and learning. If astronomy interests you, then one program that will let you browse through the infinity of space without having to go outdoors is Stellarium. It's free, open-source and available for download from www.stellarium.org.

In this article, we'll be looking at how to use Stellarium and we'll also take a panoramic photo of our surroundings and then use it in the application itself. Why so? This lets you use the surroundings you are in and get the exact location of the planets, nebulae, stars and other objects in the sky. It makes pinpointing objects in the sky much simpler and it informs you in advance when and where a planet or star will appear along your actual horizon.

A look at the interface and how to use it

Stellarium's interface is extremely simple and it runs mostly as a full-screen application. The software shows you a perspective view of the sky standing from a point on the Earth. You can change this point of view whenever you wish.

There are two main toolbars on the left bottom of the screen. The horizontal one contains all the toggling for the elements in the scene. This basically allows you to hide away the trees and the ground or turn on the constellations and the artwork. You can even turn off the atmosphere.

The toolbar on the left contains the application controls. Here, the current time and location from where you are viewing the sky can be set. The search function is also available to find a particular object in the sky.

Navigating through space is simple. Use the arrow keys to look around. Think of yourself looking up, down and sideways. To zoom in, use the Page Up and Page Down keys. You can also use the mouse-scroll. Left-click to select an object and to view information on it which is displayed on the left top. To hide the information, right-click.

Setting time and starting real-time

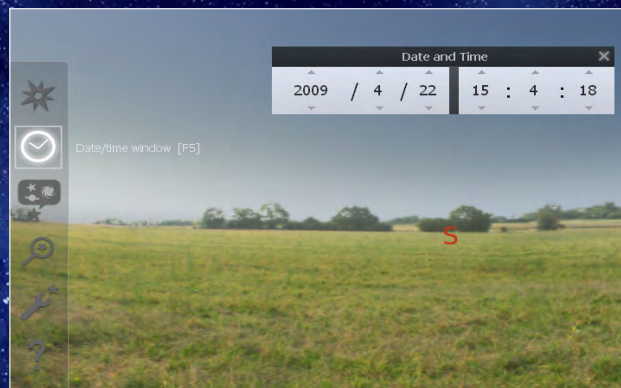
In Stellarium, just like in real-life, the skies can only be viewed at night. The only way to see the skies during the daytime is to turn off the environment. This can be done by clicking the **Atmosphere** button on the horizontal toolbar on the bottom of the screen.

What is a panorama and why do we need one?

A panorama is basically a single image that you could wrap around virtually so as to give you the freedom to look all around you. A panorama is great because you can use it to relate to different points in the sky. For example, you can be looking up to the sky and can expect a star to rise from behind a building in your vicinity well in advance. To look for that same star otherwise would be a more confusing task.

How to shoot a panorama?

Shooting a panorama shot is the difficult part of creating a panorama, at least in comparison to the stitching of the images. In short, a whole sequence of photos needs to be shot of the area you plan on using as a panorama. The photos have to be taken in the open. This sequence of photos must cover the entire area of which you wish to create a panorama. Start from a position at eye level and keeping



Setting the current date and time

the zoom to a minimum, start clicking photos while keeping in mind the area covered in the viewport. Next, move to the adjacent section of the view and click another photo. Continue doing this for all the photos in a row and shoot multiple rows till you have covered the entire 360 degrees, including the area above you.

While taking the photos, try not to change any of the settings on the camera. This ensures that there will be no big changes in the colours, brightness and contrast of the images. If you have a camera with manual settings, switch to that mode and set a comfortable setting for the ISO, aperture, shutter speed and megapixels. Use settings such that the scene is well illuminated and the shutter speed is kept reasonably fast so as to avoid any blurring.

You should also do this exercise really quickly so you don't have a big difference in lighting. If you start in the morning and end in the evening, there's a big chance that one side of your panorama is going to be dull or grainy. Be prepared with a charged set of batteries and a large memory card as well. As the end result is going to be a large image, it's alright to turn down the sensor resolution to something small, such as 1600 x 1200.

Stitching it up

The panorama sequence of shots cannot be used separately but they have to be merged together into one. This can be done with a couple of programs. Using Adobe's Photoshop CS3 is one of the simpler ways of getting a panorama shot done. Firstly, dump the sequence of images into one folder on your hard drive.

Start Photoshop, then go to **File > Automate > Photomerge**. Set the layout as **Auto** and select the files or folder that has the images. Click **OK** when done. (See figure 1). Photoshop will then show you a preview of the panorama. (See figure 2) Photoshop will then load the images, align them and merge them into a single file. Save the image when you're done.

Editing the panorama image

Now that the panorama is created, we need to clear out the sky so that the skies can be viewed. Double click on the only layer that exists - the background and make it a layer. Now, use the Eraser Tool and then erase the area above the horizon. To be even neater, use the line tool from the toolbar on the left. Draw a line and make as many nodes as required.